EEPROM

FT24C32A EEPROM Module

v1.0

2025-09-05

Rev. A

Professional electronic component

PRODUCT OVERVIEW

EEPROM module based on the FT24C32A chip, providing 32Kbits of non-volatile memory storage via I2C interface. Ideal for applications requiring reliable data retention and easy integration with microcontrollers.

KEY TECHNICAL SPECIFICATIONS

🔌 CONNECTIVITY

Primary Interface:	**GPIO (Interrupt)**
Connector Type:	**JST 4-pin 1.0mm**
Logic Levels:	**VCC-referenced (2V – 5.5V tolerant)**

PIN CONFIGURATION

FUNCTION	NOTES
`Power Supply`	3.3V or 5V
`Ground`	Common ground for all components

KEY FEATURES

Integrated Chip:
FT24C32A 32Kbit I2C EEPROM

Interface:
I2C (2-wire)

Operating Voltage:
1.8V to 5.5V

Communication Speed:
Up to 400 kHz

ADDITIONAL TECHNICAL INFORMATION

BASIC COMMANDS

COMMAND	DESCRIPTION	EXAMPLE
`scan`	Scan for I2C devices and detect EEPROM	`scan`
`read `	Read data from memory address	`read 0x0000 16`
`write `	Write text to memory address	`write 0x0000 Hello World`
`erase`	Erase entire EEPROM (fill with 0xFF)	`erase`
`wpcheck [addr]`	Check write protection status	`wpcheck 0x0000`
`addr `	Set EEPROM I2C address manually	`addr 0x50`

EEPROM SPECIFICATIONS

PARAMETER	VALUE
Capacity	32 Kbits (4,096 bytes)
Interface	I2C (2-wire)
I2C Address	0x50 - 0x57 (A0, A1, A2 configurable)
Page Size	32 bytes
Operating Voltage	1.8V - 5.5V
Max Clock Speed	400 kHz
Write Cycle Time	≤ 5ms
Data Retention	> 40 years
Endurance	> 1,000,000 cycles

TYPICAL APPLICATIONS

APPLICATION	DESCRIPTION
Data logging and storage	Store sensor readings, logs, or event data for later retrieval.
Configuration storage	Save device settings or calibration data that must persist after power loss.
Non-volatile memory for microcontroller projects	Provide reliable memory for storing critical information in embedded systems.

HARDWARE DOCUMENTATION

MECHANICAL DIMENSIONS

Dimensions

Physical dimensions and mounting specifications (measurements in millimeters)

SYSTEM TOPOLOGY

System Topology

Connection topology and system integration diagram

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COMPONENT REFERENCE

REF.	DESCRIPTION
IC1	{{sensor_description}}
L1	Power On LED
U1	{{regulator_description}}
JP1	2.54 mm Castellated Holes
J1	QWIIC Connector (JST 1 mm pitch) for I2C

ISSUE	SOLUTION
EEPROM not detected	Check wiring, I2C address, and pull-up resistors
Write errors	Verify WP pin is not high, check power supply
Communication errors	Reduce I2C clock speed, check connections
Data corruption	Ensure proper timing between operations

CIRCUIT SCHEMATIC

Circuit Schematic

Complete circuit schematic showing all component connections

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION	NOTES
Power Supply	3.3V or 5V
Ground	Common ground for all components

VOLTAGE LEVEL	FUNCTION
3.3 V – 5.5 V	Provides power to the on-board regulator and sensor core.
0 V	Common reference for power and signals.
1.8 V to VCC	Serial data line for I²C communications.
1.8 V to VCC	Serial clock line for I²C communications.

PIN CONFIGURATION LAYOUT

Physical connector layout and pin positioning

Pin Configuration Layout

Complete pin configuration diagram showing all connectors, pin assignments, and electrical connections for proper integration

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